DSA03-Natural Language Processing

List of Lab Program

2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we will create a simple automaton to match strings ending with 'ab' using python.

Code:

```
def is_accepted(string):
    state = 0
    for ch in string:
        if state == 0:
            if ch == 'a':
                state = 1
            else:
                state = 0
        elif state == 1:
            if ch == 'a':
                state = 1
            elif ch == 'b':
                state = 2
        elif state == 2:
            if ch == 'a':
                state = 1
            else:
                state = 0
```

```
    return state == 2
string = input("Enter a string: ")
if is_accepted(string):
    print("Accepted")
else:
    print("Rejected")
```

Output:



